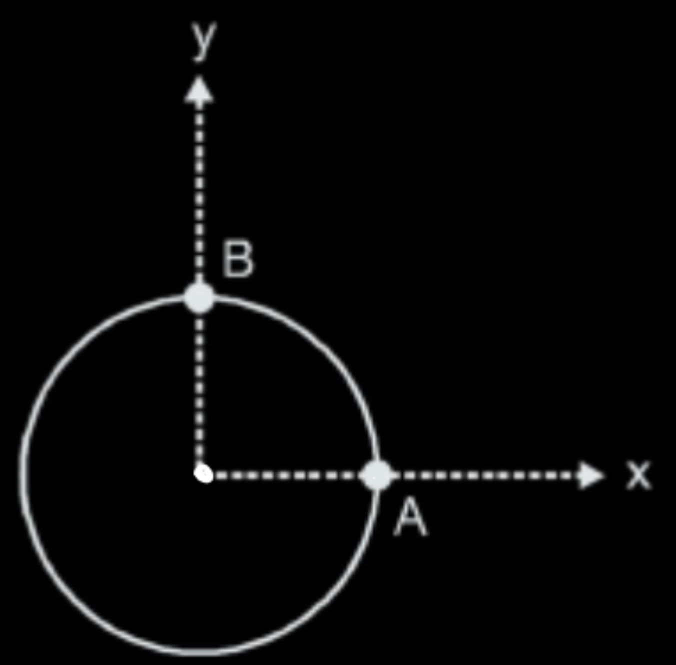
A uniform disc of mass m and radius R, hinged at some point is performing pure rotation with respect to hinge, in horizontal plane with a constant angular velocity w. At certain instant center of the disc is at origin in the mentioned co-ordinate system and velocity of particle A is $\vec{V}_A = -\frac{\omega R}{4}(3\hat{i} - 4\hat{j})$ m/s. Kinetic energy of the disc is –

$= \frac{\omega R}{4}(-3\hat{i} + 4\hat{j})$ m/s

$K.E. = \frac{1}{2} I_H \omega^2$

$v_A = \frac{\omega R}{4} \times 5 = \omega\left(\frac{5R}{4}\right) = \omega r$



(A) $\frac{mR^2\omega^2}{2}$

(B) $\frac{17mR^2\omega^2}{4}$

(C) $\frac{17mR^2\omega^2}{32}$

(D) $\frac{7mR^2\omega^2}{32}$

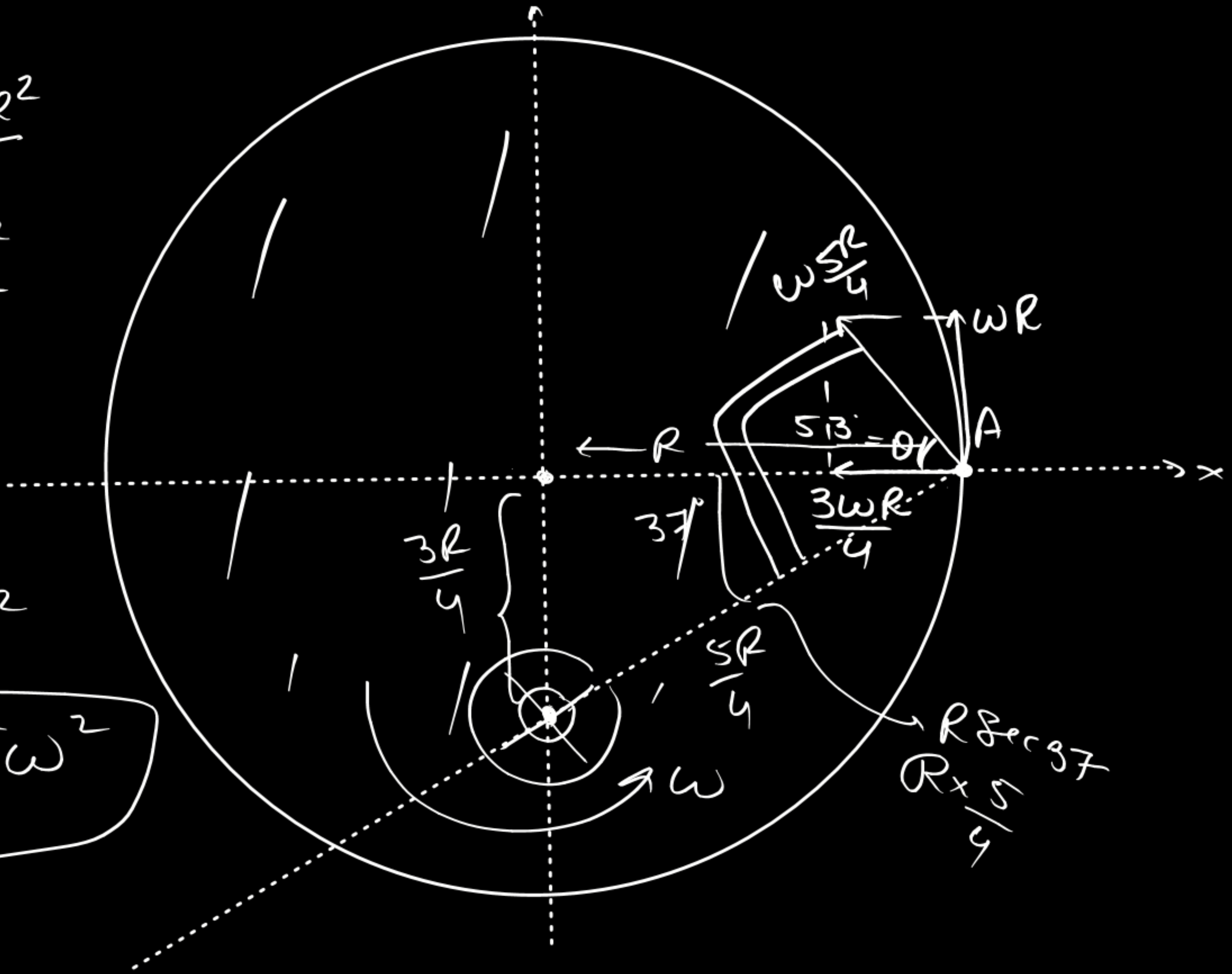
$$I_M = \frac{mR^2}{2} + m\frac{9R^2}{16}$$

$$= \frac{(8+9)mR^2}{16}$$

$$= \frac{17mR^2}{16}$$

$$\frac{1}{2}\cdot\frac{17mR^2}{16}\omega^2$$

$$\boxed{\frac{17mR^2\omega^2}{32}}$$



One mole of monoatomic ideal gas is initially at pressure P_0 and volume V_0. The gas then undergoes a three-stage cycle consisting of the following processes:

(i) An isothermal expansion till it reaches volume $2V_0$, and heat Q flows into the gas

(ii) An isobaric compression back to the original volume V_0

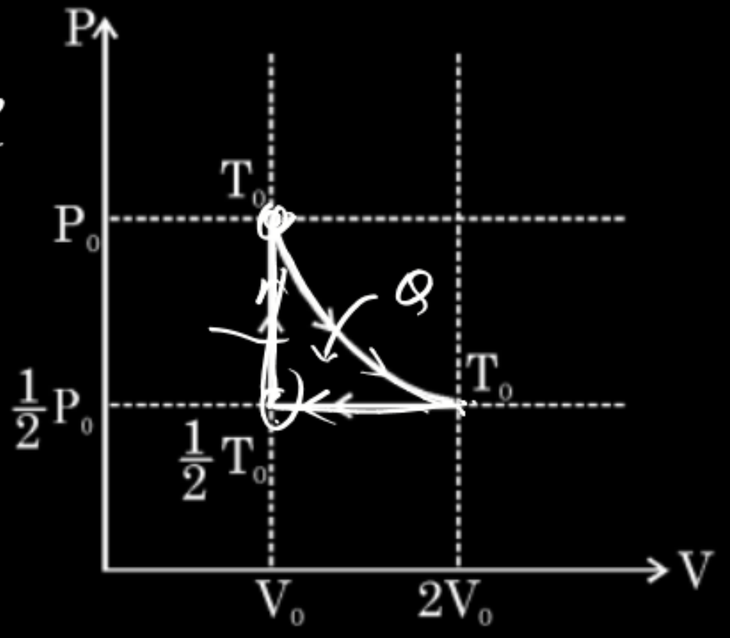
(iii) An isochoric increase in pressure till the original pressure P_0 is regained.

The efficiency of this cycle can be expressed as

$n = 1$

$f = 3$

$\eta = \dfrac{W}{Q_{in}}$



The corresponding P-V diagram is shown above.

(i) $\Delta U = 0 \Rightarrow W = \Delta Q = Q$

(ii) $W = P\Delta V = \dfrac{P_0}{2}(-V_0) = \dfrac{-P_0V_0}{2} = \dfrac{-RT_0}{2}$

$\Delta Q = \Delta U + W$

$= nC_p\Delta T$

$= 1 \times \dfrac{5}{2}R\left(-\dfrac{T_0}{2}\right) = -\dfrac{5}{4}RT_0$

(iii) $W = 0$

$\Delta Q = nC_v\Delta T$

$= 1 \times \dfrac{3}{2}R \times \dfrac{T_0}{2} = \dfrac{3RT_0}{4}$

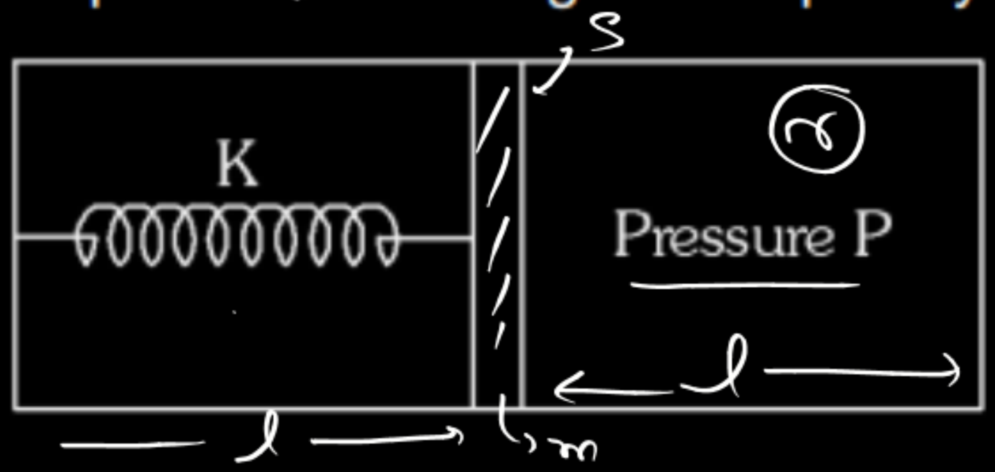
$P_0V_0 = 1 \times RT_0$

$\eta = \dfrac{Q - \frac{RT_0}{2} + 0}{Q + \frac{3RT_0}{4}}$

$= \dfrac{4Q - 2RT_0}{4Q + 3RT_0}$

(A) $\epsilon = \dfrac{4Q - 2RT_0}{4Q + 3RT_0}$ (B) $\epsilon = \dfrac{4Q + 2RT_0}{4Q - 3RT_0}$ (C) $\epsilon = \dfrac{4Q - 2RT_0}{4Q + RT_0}$ (D) $\epsilon = \dfrac{4Q + 2RT_0}{4Q + RT_0}$

A non–conducting piston of mass m and area S divides a non-conducting, closed cylinder into two parts as shown in Figure. Piston is connected with left wall of cylinder by a spring of force constant K. Left part is evacuated and right part contains an ideal gas at pressure P. Adiabatic constant of the gas is γ and in equilibrium length of each part is ℓ. The angular frequency of small oscillation of piston is

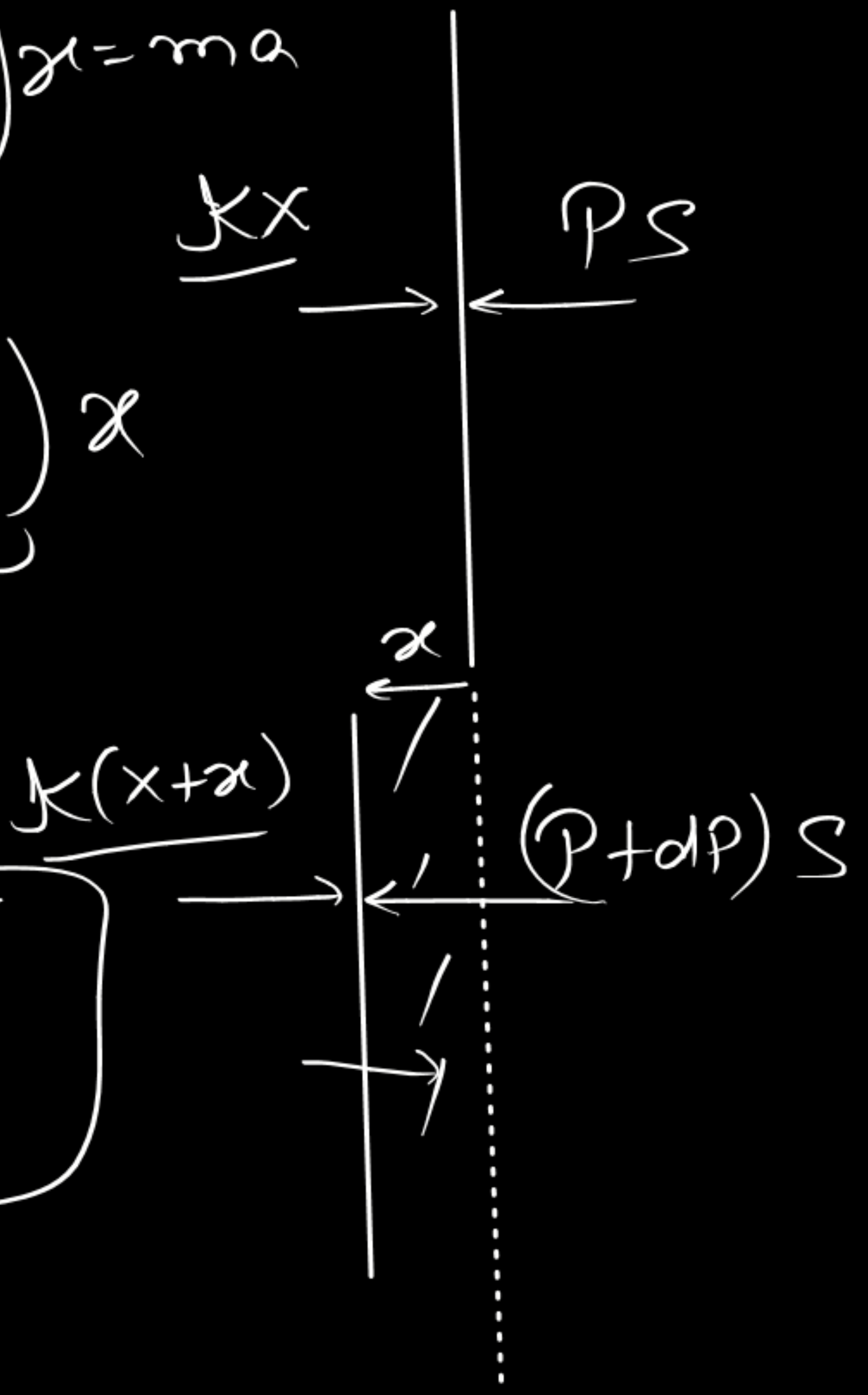


(A) $\sqrt{\frac{2K\ell+\gamma PS}{m\ell}}$ (B) $\sqrt{\frac{K\ell+\gamma PS}{2m\ell}}$ (C) $\sqrt{\frac{2(K\ell+\gamma PS)}{m\ell}}$ (D) $\sqrt{\frac{(K\ell+\gamma PS)}{m\ell}}$

$$f = \left(k + \frac{\gamma P S}{l}\right)x = ma$$

$$a = \underbrace{\left(\frac{k}{m} + \frac{\gamma P S}{m l}\right)}_{\omega^2} x$$

$$\boxed{\omega = \sqrt{\frac{k l + \gamma P S}{m l}}}$$

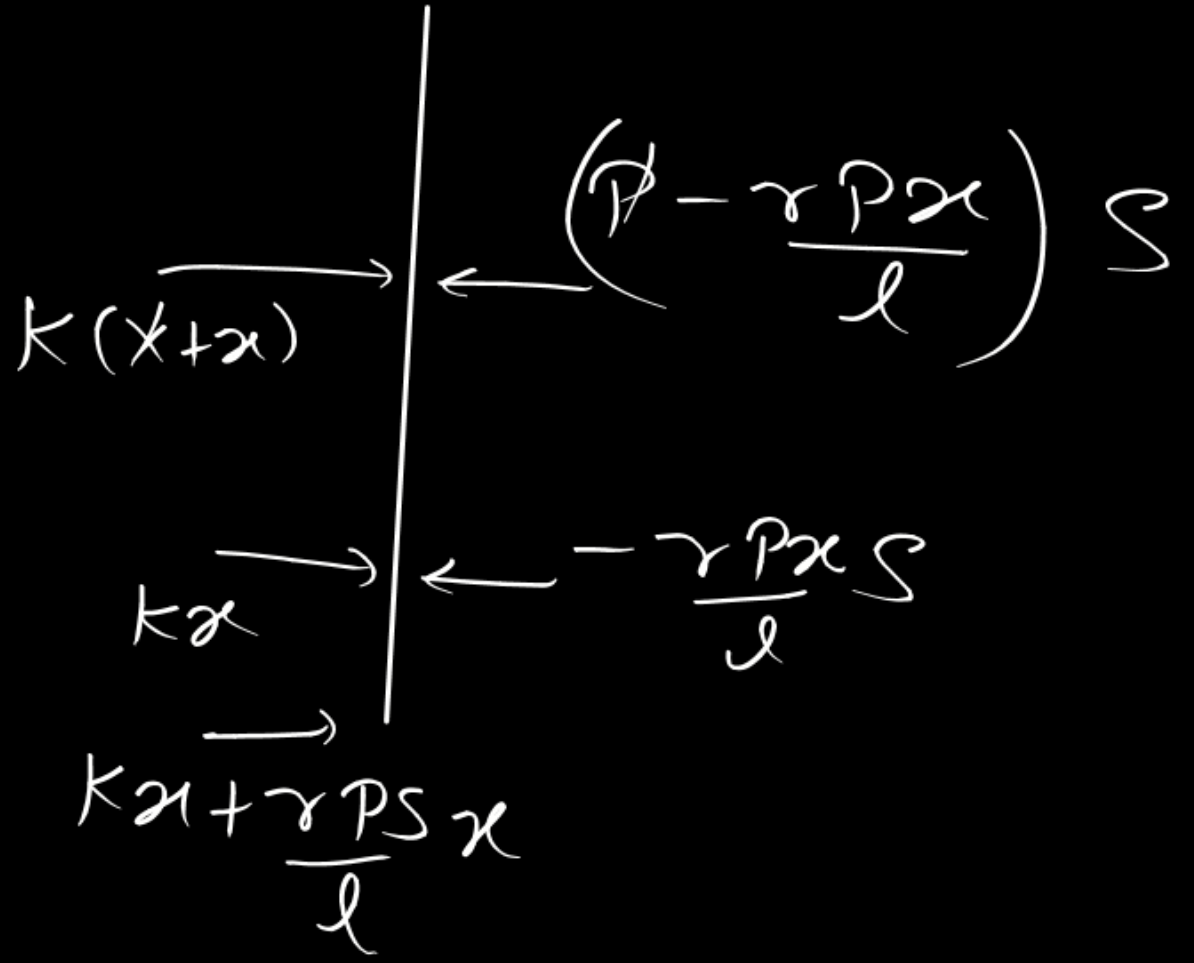


$$PV^\gamma = \text{const}$$

$$\boxed{PS = kX}$$

$$\frac{dP}{P} = -\gamma \frac{dV}{V}$$

$$dP = -\gamma P \frac{dV}{V} = -\gamma P \cdot \frac{\cancel{S} x}{\cancel{S} l}$$



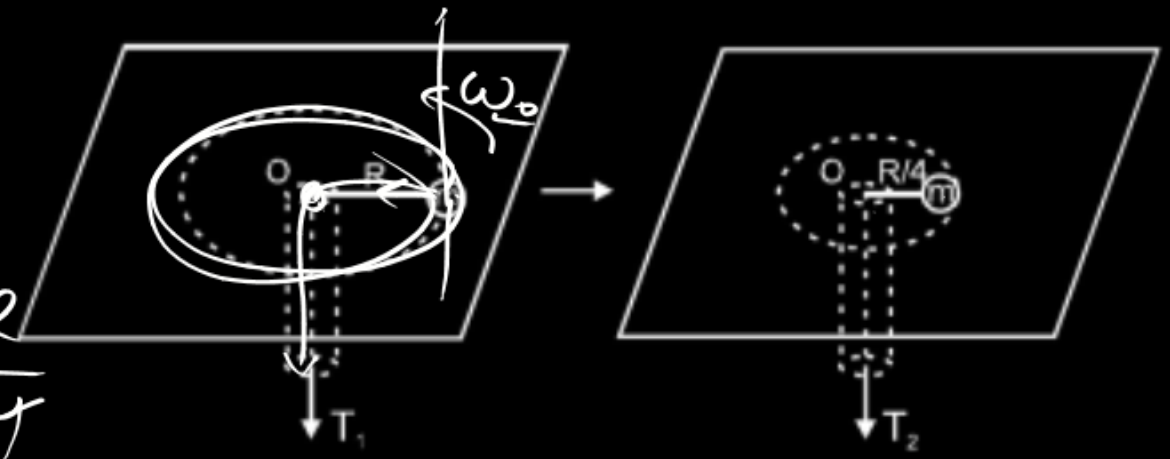
A ball of mass m is connected by a light inextensible cord and is rotated in a circle of radius R on a smooth fixed horizontal table. Initially the angular velocity of the ball was ω_0 and pulling force applied was T_1. Now the pulling force is increased to T_2, until the radius of rotation of the ball becomes $\frac{R}{4}$.

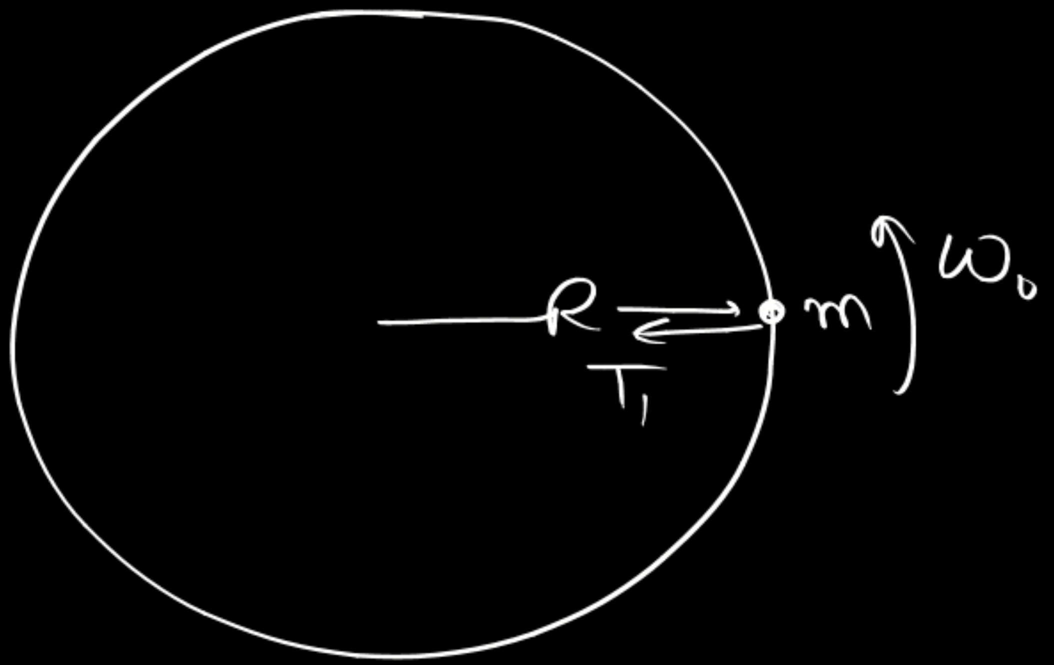
Then ratio $\frac{T_2}{T_1}$ is :

$\omega_0 \times 16$

$R/4$

$T_2 = m\,\cancel{256}\,64\,\omega_0^2 \frac{R}{4}$





(A) $\frac{1}{4}$ (B) 4 (C) 16 (D) 64

$\boxed{T_1 = m\omega_0^2 R}$

$T_2 = 64 m\omega_0^2 R$

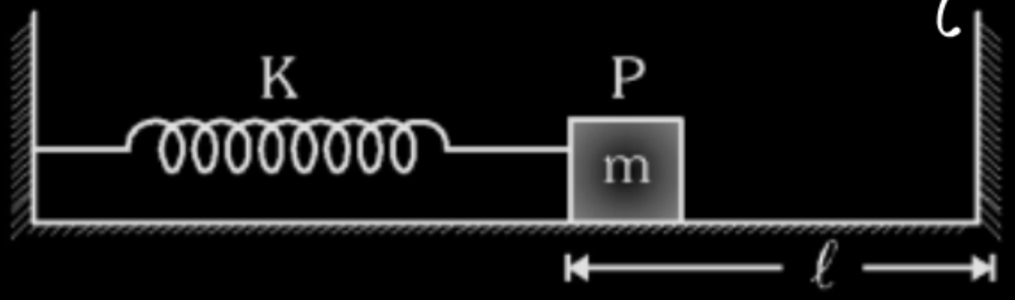
$T_2 = 64 T_1$

$m(\omega_0 R)R = m\left(\omega \frac{R}{4}\right)\frac{R}{4}$

$\boxed{\omega = 16\omega_0}$

Figure shows a block P of mass m resting on a smooth horizontal surface, attached to a spring of force constant k which is rigidily fixed oin the wall on left side. At a distance ℓ to the right of block there is a rigid wall. If block is pushed toward left so that spring is compressed by a distance $\frac{5\ell}{3}$ and released, it will start its oscillations. If collision of block with the wall is considered to be perfectly elastic. The time period of oscillation of the block is

$e=1$



Figure

(A) $\sqrt{\frac{m}{K}}\left[\pi+\cos^{-1}\left(\frac{3}{5}\right)\right]$

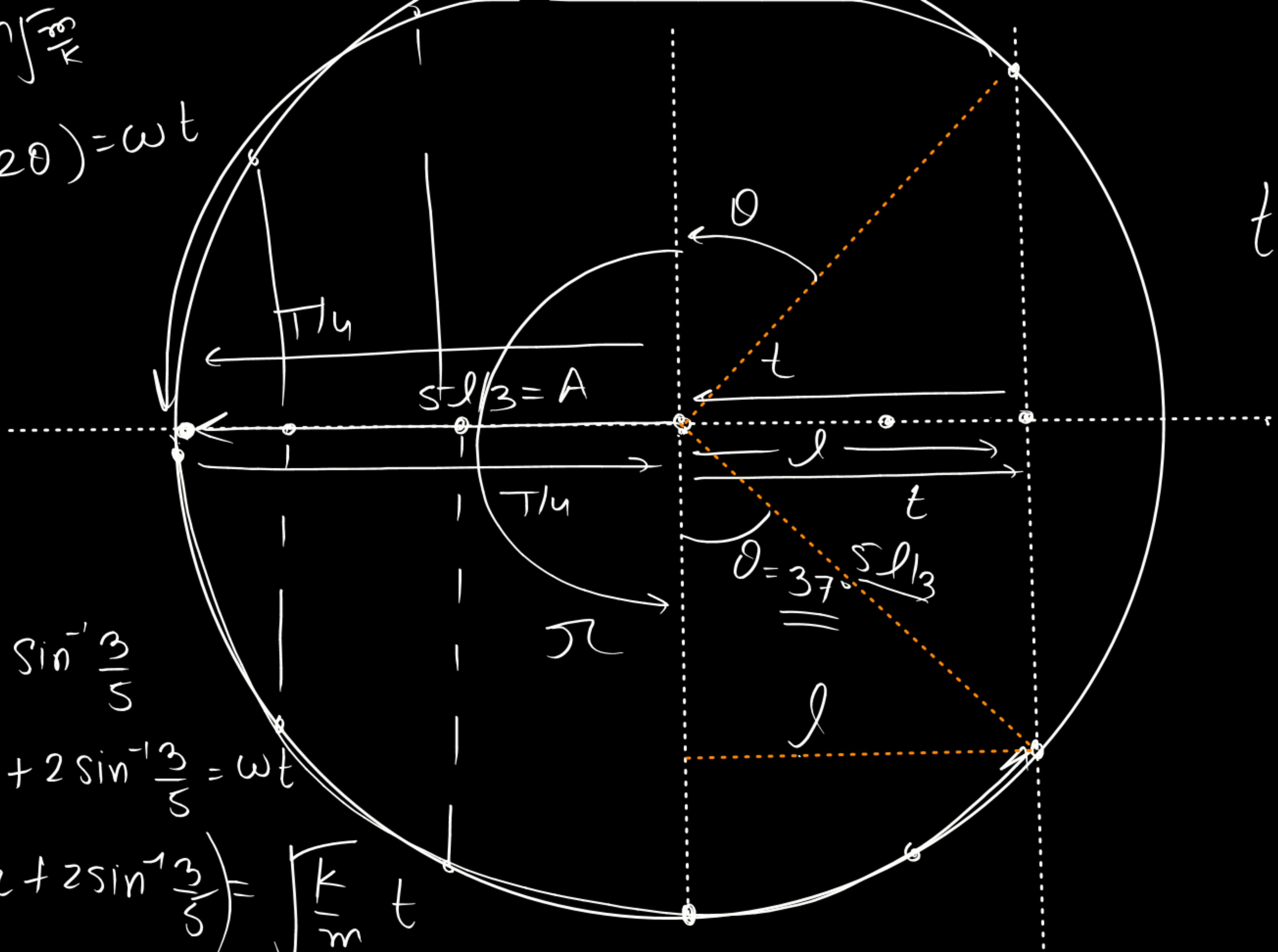
(B) $\sqrt{\frac{m}{K}}\left[\pi+2\cos^{-1}\left(\frac{3}{5}\right)\right]$

(C) $\sqrt{\frac{m}{K}}\left[\pi+2\sin^{-1}\left(\frac{3}{5}\right)\right]$

(D) $\sqrt{\frac{m}{K}}\left[\pi+\sin^{-1}\left(\frac{3}{5}\right)\right]$

$T = 2\pi\sqrt{\frac{m}{k}}$

$(\pi + 2\theta) = \omega t$



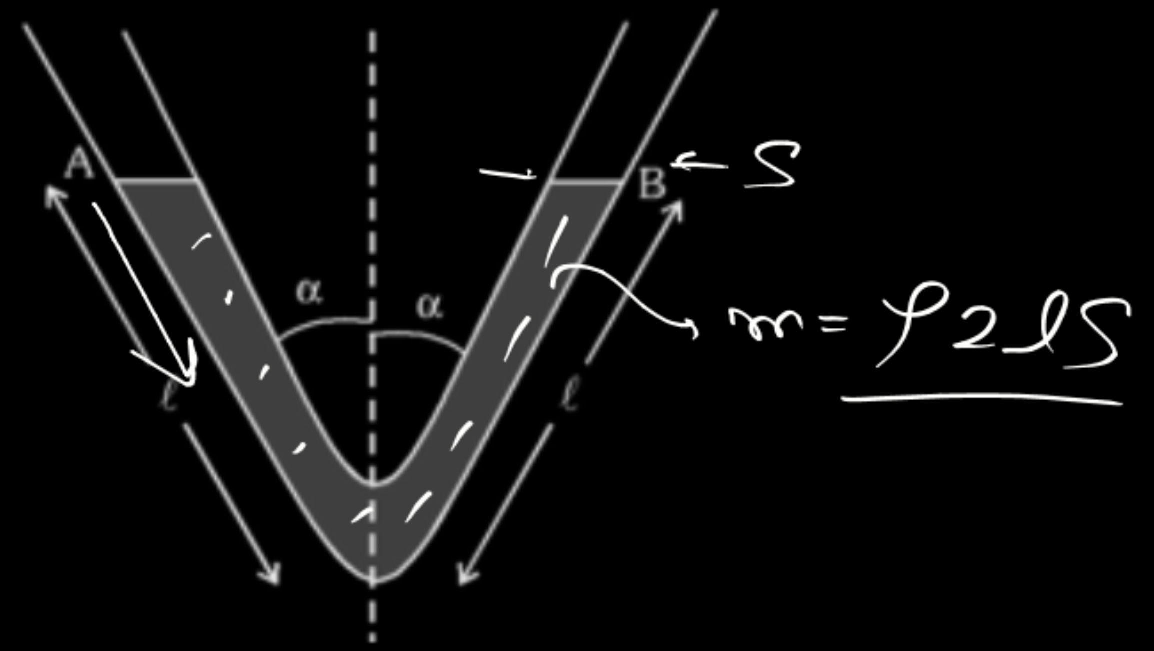
$t = \sqrt{\frac{m}{k}}\left(\pi + 2\sin^{-1}\frac{3}{5}\right)$

$\theta = \sin^{-1}\frac{3}{5}$

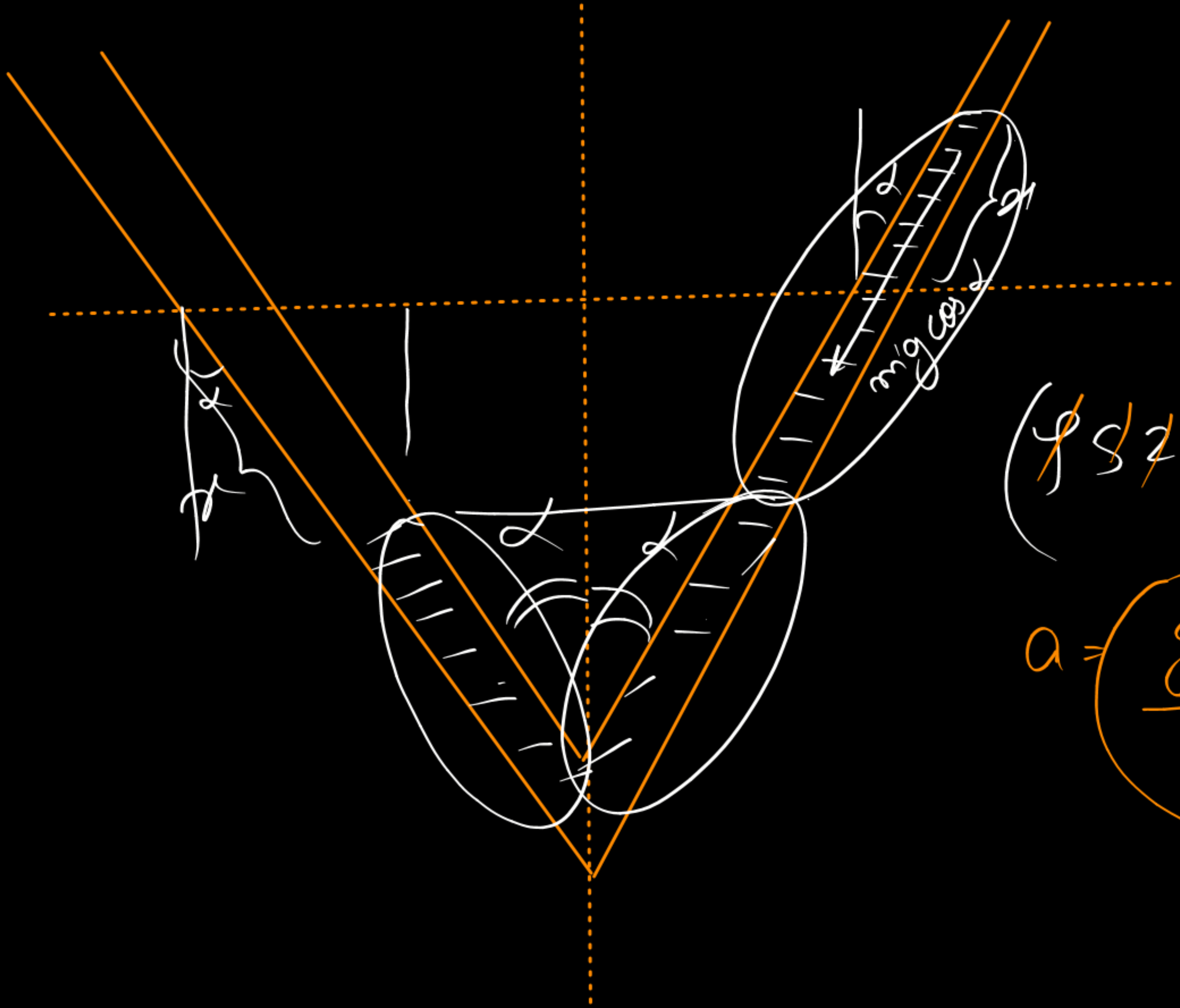
$\pi + 2\sin^{-1}\frac{3}{5} = \omega t$

$\left(\pi + 2\sin^{-1}\frac{3}{5}\right) = \sqrt{\frac{k}{m}}\, t$

A V-tube of uniform cross–section area A is filled with a liquid of density p, as shown in figure. Each arm has length ℓ and inclined to vertical at an angle α. If the free surface of the liquid in one arm is disturbed by displacing it a depth x below the equilibrium configuration, the frequency of oscillation is



(A) $f = \frac{1}{2\pi}\sqrt{\left(\frac{g\sin\alpha}{\ell}\right)}$ (B) $f = \frac{1}{2\pi}\sqrt{\left(\frac{g\cos\alpha}{\ell}\right)}$ (C) $f = \frac{1}{2\pi}\sqrt{\left(\frac{2g\cos\alpha}{\ell}\right)}$ (D) $f = \frac{1}{2\pi}\sqrt{\left(\frac{2g\sin\alpha}{\ell}\right)}$



$$\left(\cancel{\rho}\cancel{S}\cancel{2}x\,g\cos\alpha\right) = \left(\cancel{\rho}\cancel{S}\cancel{2}l\right) a$$

$$a = \frac{g\cos\alpha}{l}\,x \qquad \omega^2$$

$$\omega = \sqrt{\frac{g\cos\alpha}{l}}$$

$$\boxed{f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g\cos\alpha}{l}}}$$

Two blocks of mass m_1 and m_2 ($m_1 < m_2$) are connected with an ideal spring on a smooth horizontal surface as shown in figure. At $t = 0$ m_1 is at rest and m_2 is given a velocity v towards right. At this moment, spring is in its natural length. Then choose the correct alternative :

$m_2 v + 0 = \underline{m_2 v}$

$K.E_i = \frac{1}{2} m_2 v^2 = T.E$

$\underline{v' > v}$

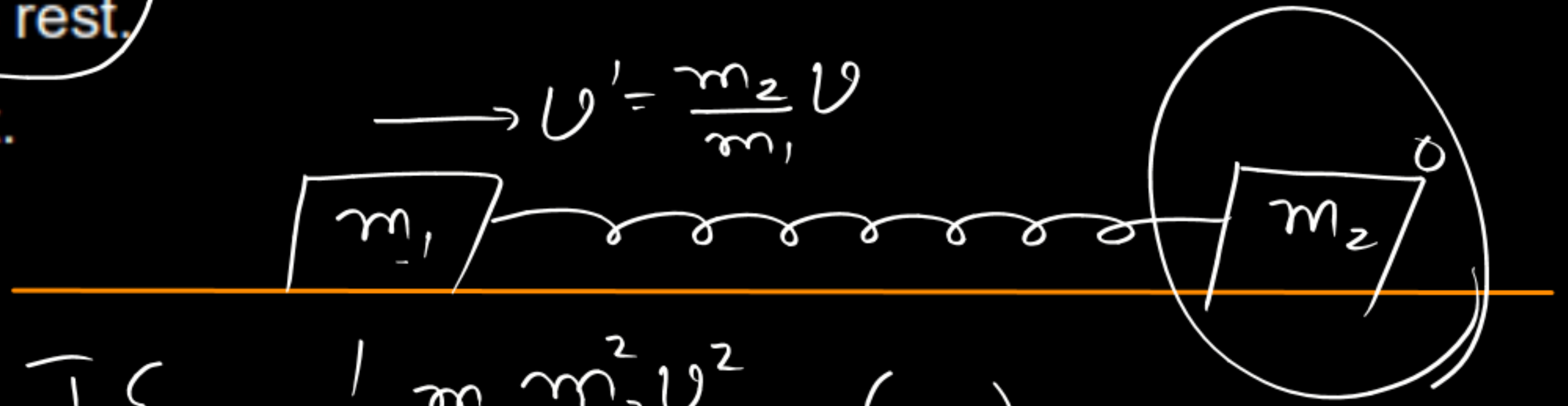
(A) Block of mass m_2 will be finally at rest after some time.

(B) Block of mass m_2 will never come to rest.

(C) Both the blocks will be finally at rest.

(D) None of these

$\longrightarrow v' = \frac{m_2 v}{m_1}$



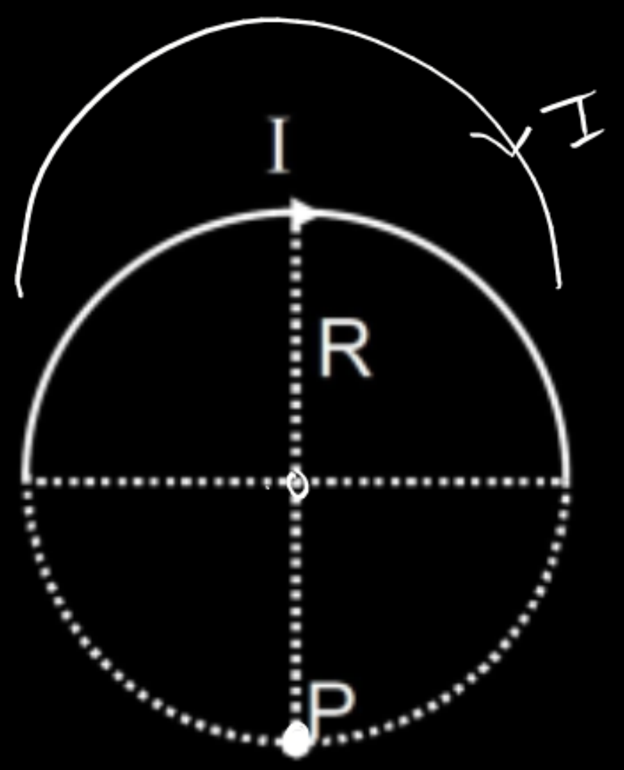
$m_2 v + 0 = m_1 v' + 0$

$v' = \frac{m_2}{m_1} v$

$T.E. \quad \frac{1}{2} m_1 \frac{m_2^2 v^2}{m_1^2} + (\quad)$

$\underline{T.E.} = \frac{1}{2} \frac{m_2^2}{m_1} v^2 + (\quad) = \frac{1}{2} m_2 v^2 \left(\frac{m_2}{m_1}\right) + (\quad)$

A thin wire bent into the form of a semicircle of radius R carries a current I. Magnetic field of induction at the point P as shown in the figure is:

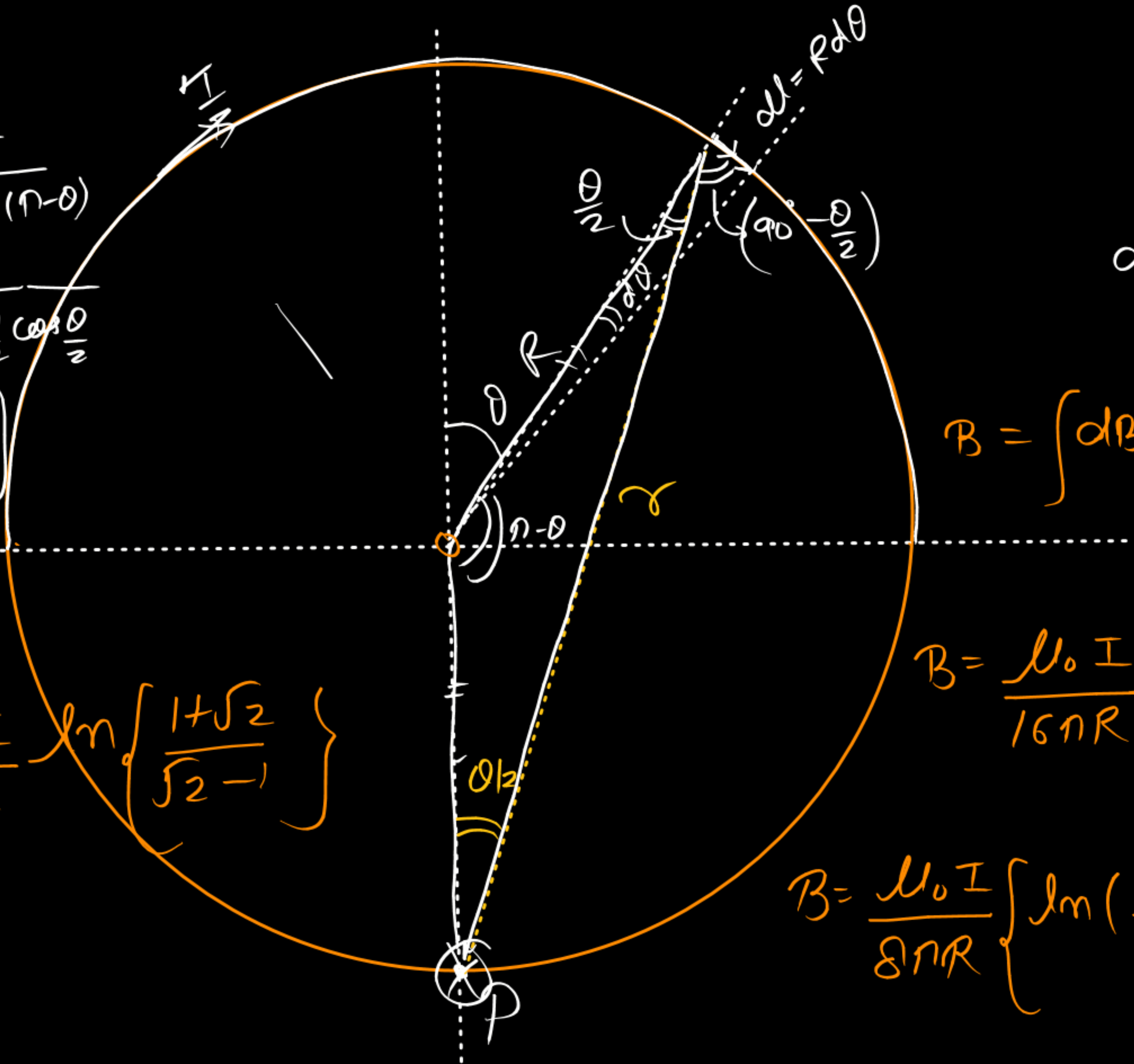


(A) $\frac{\mu_0 I}{2\pi R}\ell n\left(\sqrt{2}+1\right)$ (B) $\frac{\mu_0 I}{2\pi R}\ell n\left(\sqrt{2}-1\right)$ (C) $\frac{\mu_0 I}{4\pi R}\ell n\left(\sqrt{2}-1\right)$ (D) $\frac{\mu_0 I}{4\pi R}\ell n\left(\sqrt{2}+1\right)$

Sine law

$$\frac{R}{\sin\frac{\theta}{2}} = \frac{r}{\sin(\pi-\theta)} = \frac{r}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$

$$\boxed{r = 2R\cos\frac{\theta}{2}}$$



$$dB = \frac{\mu_0}{4\pi}\,\frac{i\,dl\sin\theta}{r^2}$$

$$dB = \frac{\mu_0}{4\pi}\,\frac{I\,\cancel{R}\,d\theta\,\sin(90-\theta/2)}{4R^{\cancel{2}}\cos^2\theta/2}$$

$$B = \int dB = \frac{\mu_0 I}{16\pi R}\int_{-\pi/2}^{\pi/2}\sec\frac{\theta}{2}\,d\theta$$

$$B = \frac{\mu_0 I}{16\pi R}\times 2\left\{\ln\left(\tan\frac{\theta}{2}+\sec\frac{\theta}{2}\right)\right\}_{-\pi/2}^{+\pi/2}$$

$$B = \frac{\mu_0 I}{8\pi R}\left\{\ln(1+\sqrt{2}) - \ln(-1+\sqrt{2})\right\}$$

$$B = \frac{\mu_0 I}{8\pi R}\ln\left\{\frac{1+\sqrt{2}}{\sqrt{2}-1}\right\}$$

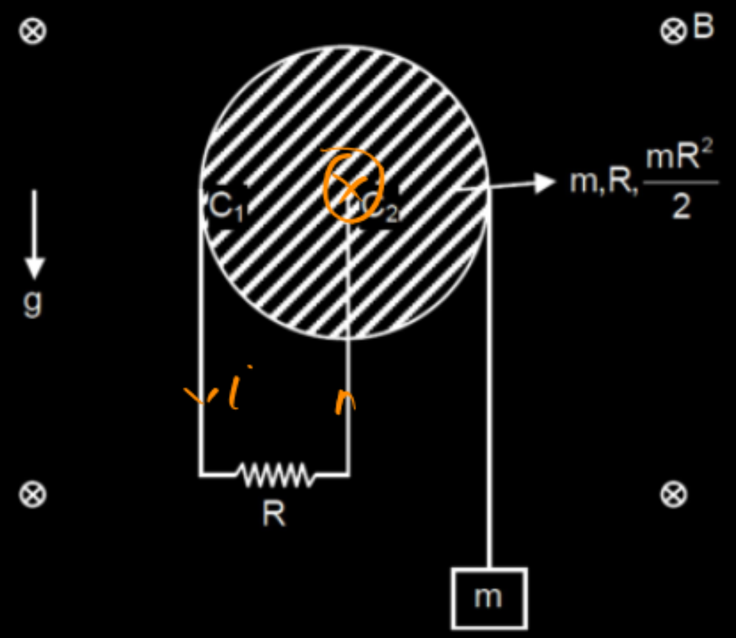
$$\frac{1+\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$\frac{(\sqrt{2}+1)^2}{2-1} = (\sqrt{2}+1)^2$$

$$B = \frac{\mu_0 I}{8\pi R} \ln(\sqrt{2}+1)^2$$

$$\boxed{B = \frac{\mu_0 I}{4\pi R} \ln(\sqrt{2}+1)}$$

Consider a perfectly conducting uniform disc of mass m and radius 'a' hinged in vertical plane from its centre and free to rotate with respect to hinge. A resistance R is connected between centre of the disc and periphery by using two sliding contacts C_1 & C_2. A long non conducting massless string is wrapped around the disc, whose another end is attached with a block of mass m. There exist a uniform horizontal magnetic field B. Whole arrangement is shown in the figure.



Given system is released from rest at t = 0. Assume friction between string and disc is sufficient so that there is no slipping between them. Let at any time t, velocity of block is v, angular velocity of disc is ω $= \frac{v}{a}$ and current in resistance is i.

(A) From the energy equation $mgv = mv\frac{dv}{dt} + I\frac{\omega d\omega}{dt} + i^2R$

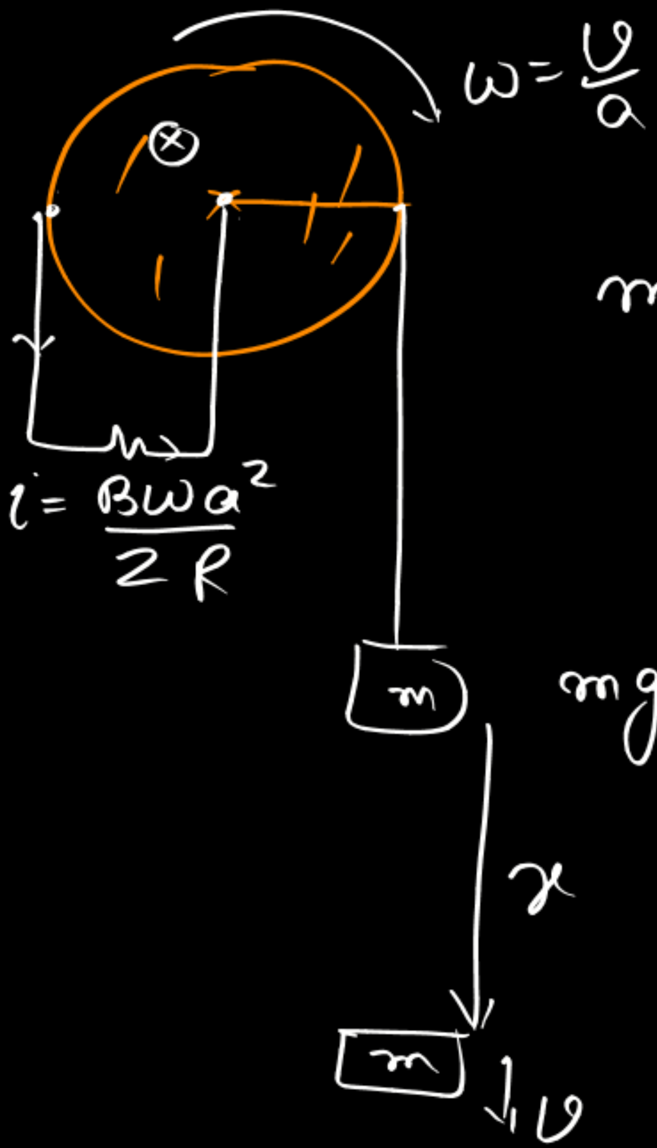
(B) The work done by the magnetic field is zero but it converts some part of the mechanical energy into heat.

(C) The velocity v, of the block as a function of time is $v = \frac{\alpha}{\beta}(1 - e^{-\beta t})$

(D) The acceleration of the block is $\frac{g}{3}e^{-\beta t}$

$a = \frac{\alpha}{\beta}\left(+\beta e^{-\beta t}\right) = \alpha e^{-\beta t}$

$\varepsilon = \frac{B\omega a^2}{2}$



$i = \frac{B\omega a^2}{2R}$

$mgx = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + \text{Heat}$

$mg\frac{dx}{dt} = \frac{1}{2}m\,2v\frac{dv}{dt} + \frac{1}{2}I\,2\omega\frac{d\omega}{dt} + i^2R$

$$\boxed{mgv = mv\frac{dv}{dt} + I\omega\frac{d\omega}{dt} + i^2R}$$

$$mgv = mv\frac{dv}{dt} + I\frac{v}{a}\frac{1}{a}\frac{dv}{dt} + i^2R$$

$$i = \frac{\mathcal{E}}{R} = \frac{B\omega a^2}{2R}$$

$$= \frac{Bva}{2R}$$

$$mg\cancel{v} = m\cancel{v}\frac{dv}{dt} + \frac{I}{a^2}\cancel{v}\frac{dv}{dt} + \frac{B^2v^{\cancel{2}}a^2}{4R}$$

$$mg - \frac{B^2a^2}{4R}v = \left(m + \frac{I}{a^2}\right)\frac{dv}{dt}$$

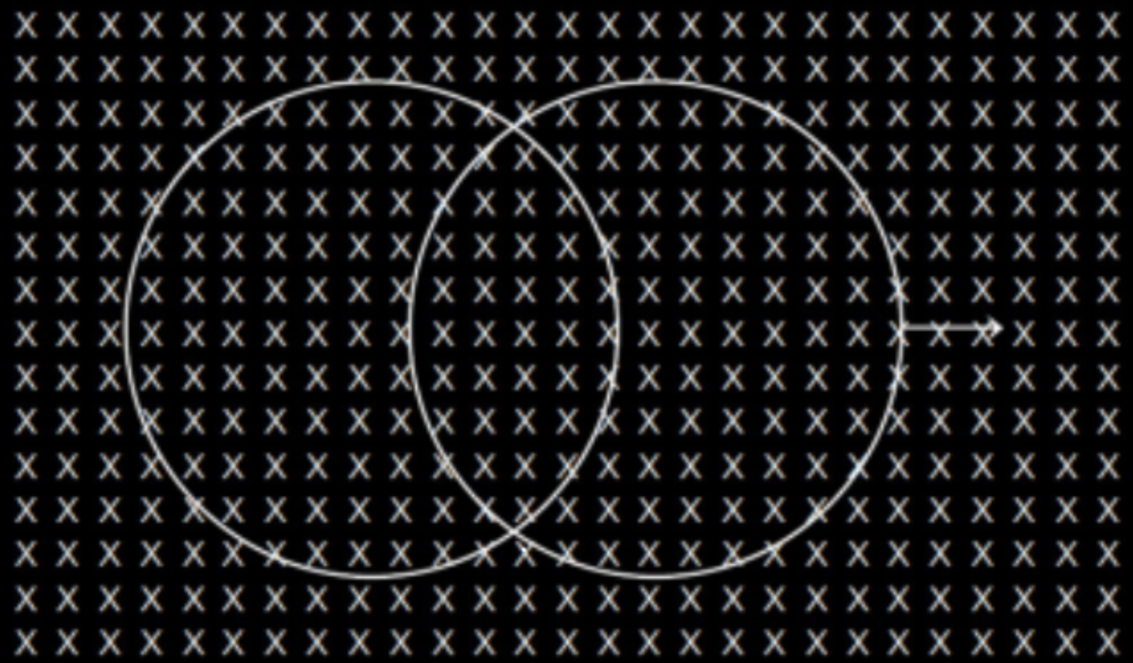
$$\int_0^t dt = \left(m + \frac{I}{a^2}\right)\int_0^v \frac{dv}{mg - \frac{B^2a^2}{4R}v}$$

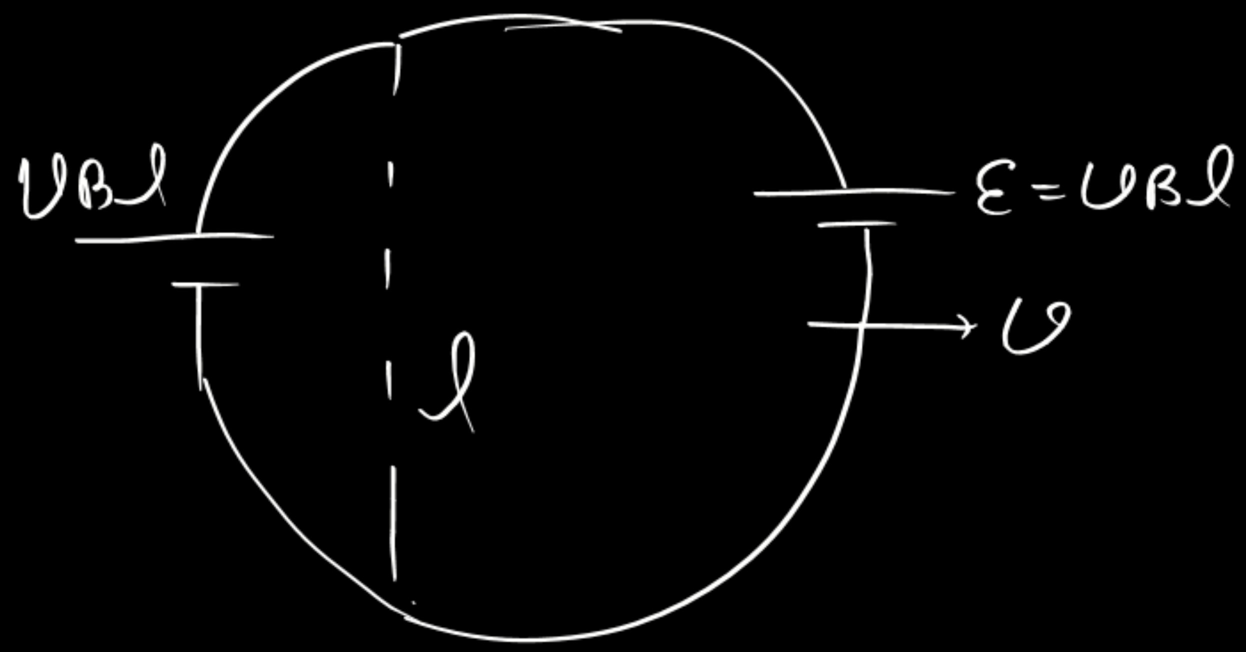
$$t = \frac{\left(m + \frac{I}{a^2}\right)}{-\frac{B^2a^2}{4R}} \ln\left(\frac{mg - \frac{B^2a^2}{4R}v}{mg}\right)$$

$$\frac{-\frac{B^2a^2}{4R}t}{\left(m + \frac{I}{a^2}\right)} = \ln\left\{1 - \frac{B^2a^2}{4mgR}v\right\}$$

$$v = \frac{4mgR}{B^2a^2}\left\{1 - e^{-(\ )t}\right\}$$

In the diagram shown, a uniform magnetic field is present perpendicular to the plane of the paper. Both the rings are identical and have a constant resistance per unit length. The left ring has been kept fixed at its position and the right ring is slid uniformly on the left ring towards the right hand side. Which of the following statements is/are true ?



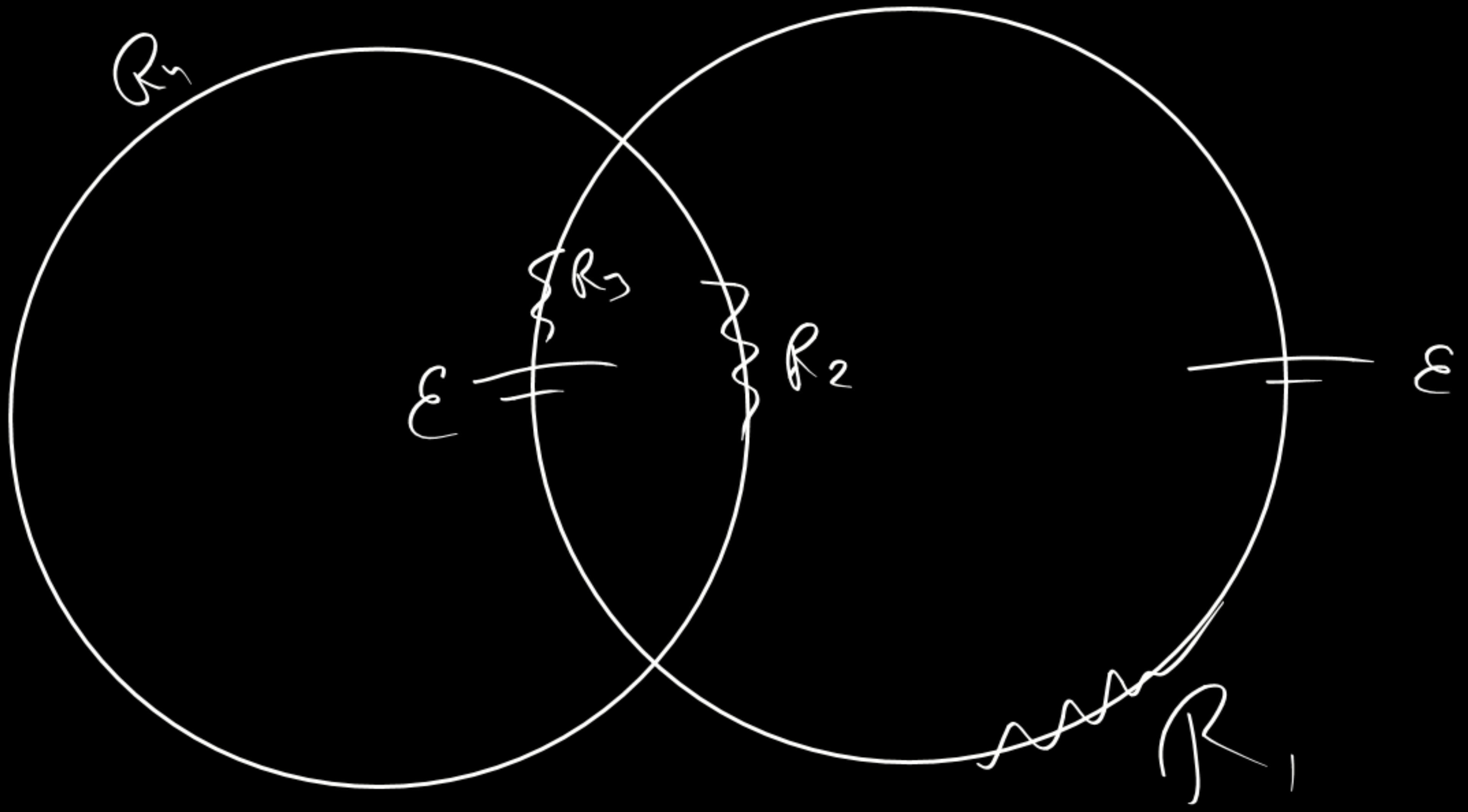


(A) The emf induced in the left ring is zero

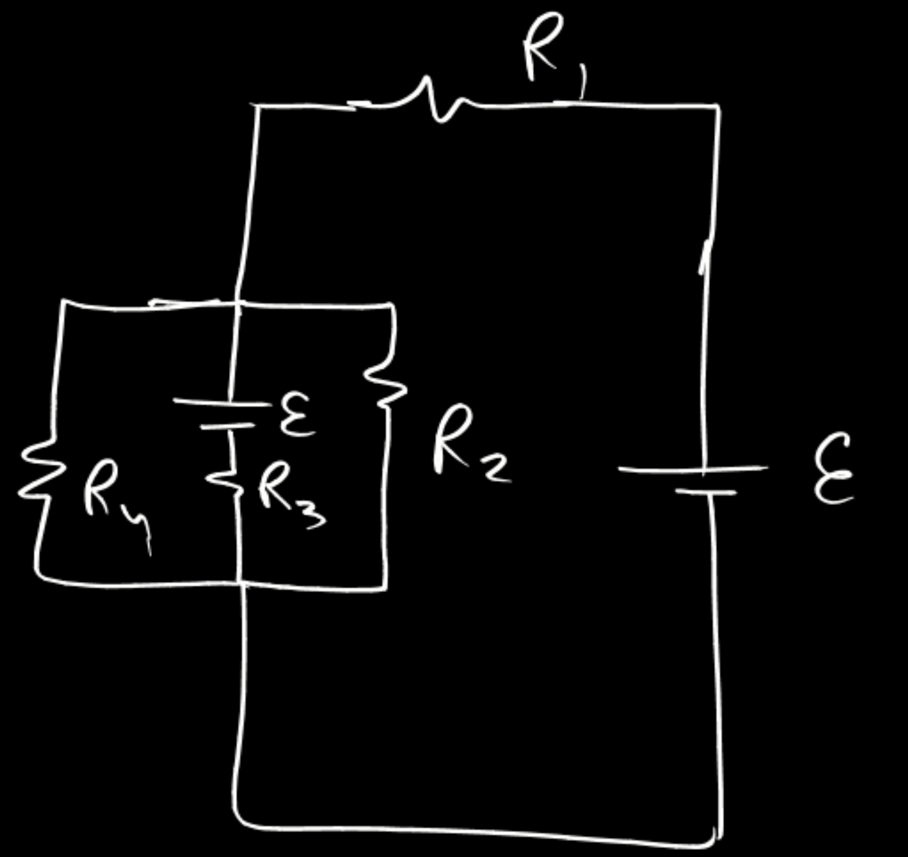
(B) The emf induced in both the rings is non zero.

(C) The magnetic force acting on the right ring is zero.

(D) The magnetic force acting on both the rings is non-zero.



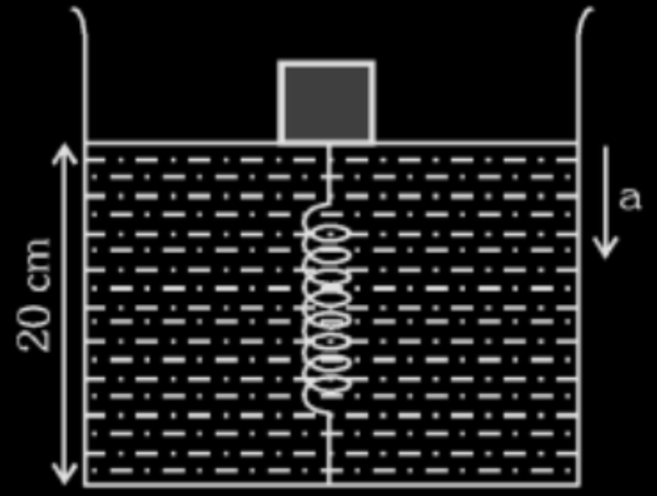
R4
R3
R2
ε
ε
R1



R1
R4
ε
R3
R2
ε

A rectangular tank having base 15 cm × 20 cm is filled with water (density ρ = 1000 kg m^{-3}) upto 20 cm height. One end of an ideal spring of natural length h_0 = 20 cm and force constant K = 280 Nm^{-1} is fixed to the bottom of the tank so that the spring remains vertical.

This system is in an elevator moving downwards with acceleration a = 2 m/s^2. A cubical block of side $\ell = 10\,\text{cm}$ and mass m = 2 kg is gently placed over the spring and released gradually, as shown in the figure as shown in the figure.



Choose the correct statement:

(A) The compression in the spring in equilibrium position is 4 cm

(B) The compression in the spring in equilibrium position is 2 cm

(C) If the block is slightly pushed down from equilibrium position and released then frequency of vertical oscillation is $\frac{5\sqrt{2}}{\pi}$ per second

(D) If the block is slightly pushed down from equilibrium position and released then frequency of vertical oscillation $\frac{3\sqrt{2}}{\pi}$ per second

The energy of the alpha particles emitted by ^{210}Po is 5.30 MeV

Given $T_{1/2}$ of ^{210}Po = 138 days.

(A) 88.4 mg mass of ^{210}Po is needed to power a thermoelectric cell of 1W output if the efficiency of energy conservation is 8%

(B) The power output after 1 year, is 0.16 W

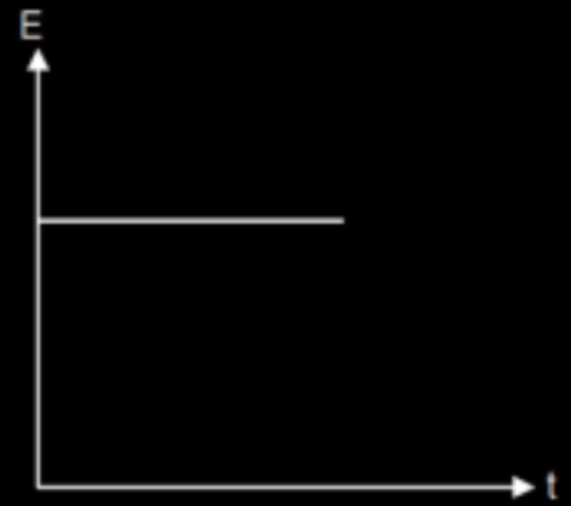
(C) 8.4 mg mass of ^{210}Po is needed to power a thermoelectric cell of 1W output if the efficiency of energy conservation is 8%

(D) The power output after 1 year, is 6.25 W

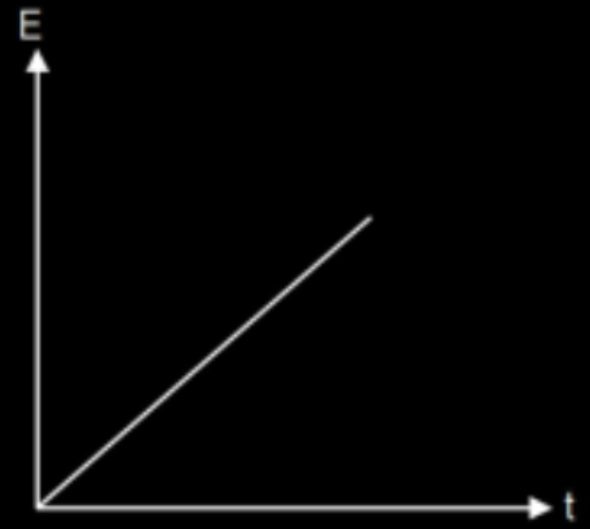
A thin non conducting ring of mass m, radius R, carrying uniformly distributed charge q is placed on smooth horizontal plane. There exist an uniform time varying magnetic field in a cylindrical region directed vertically upward. Magnitude of magnetic field varies with time as $B = B_0 t^n$ tesla, where n is a number. Centre of ring coincides with centre of cylindrical region. Ring was at rest at t = 0. Neglect the magnetic field produced due to any kind of motion of ring.

Answer the following 2 questions.

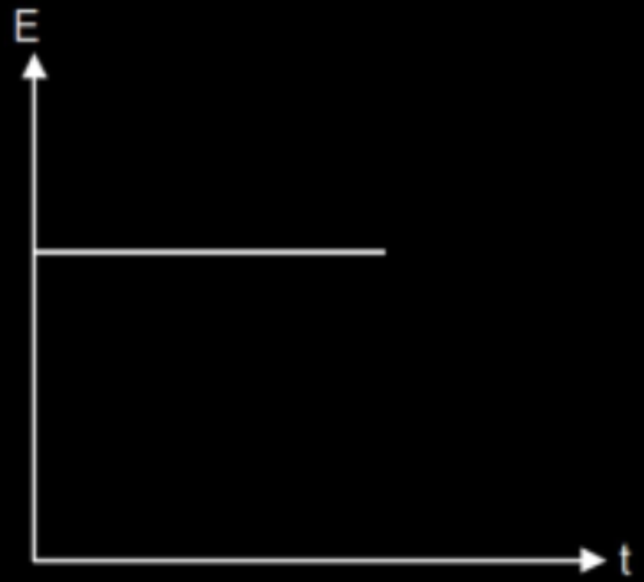
Choose the correct option(s) regarding magnitude of induced electric field at the periphery of the ring.

(A) 

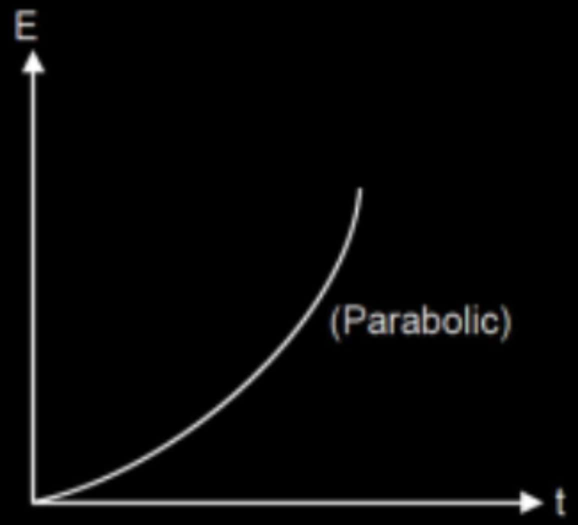
(if n = 1)

(B) 

(if n = 2)

(C) 

(if n = 2)

(D) 

(if n = 3)

A thin non conducting ring of mass m, radius R, carrying uniformly distributed charge q is placed on smooth horizontal plane. There exist an uniform time varying magnetic field in a cylindrical region directed vertically upward. Magnitude of magnetic field varies with time as $B = B_0 t^n$ tesla, where n is a number. Centre of ring coincides with centre of cylindrical region. Ring was at rest at t = 0. Neglect the magnetic field produced due to any kind of motion of ring.

Answer the following 2 questions.

Choose the correct option (s) regarding instantaneous power P delivered to the ring by the source of magnetic field.

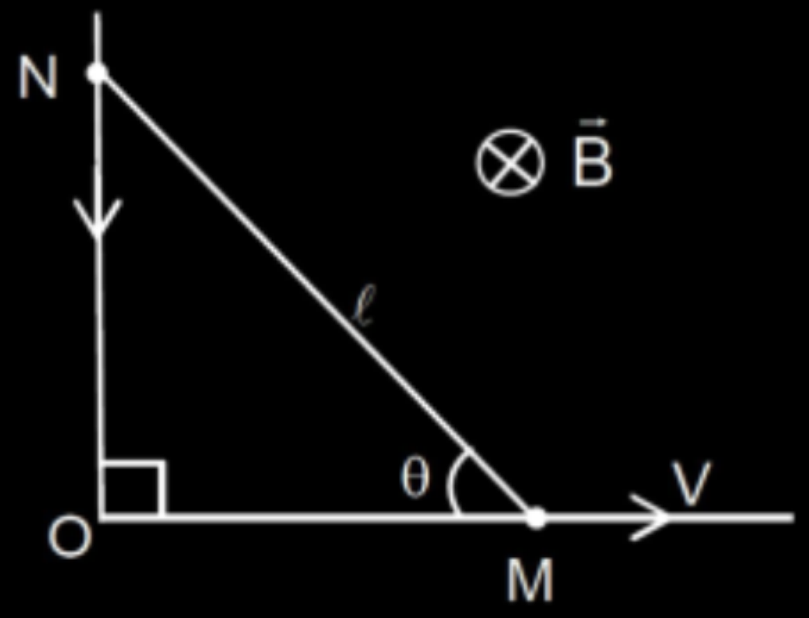
(A) if n = 1, $P = \frac{B_0^2 q^2 R^2}{4m} t$

(B) if n = 2, $P = \frac{B_0^2 q^2 R^2}{4m} t^3$

(C) if n = 4, $P = \frac{B_0^2 q^2 R^2}{m} t^7$

(D) if n = 3, $P = \frac{B_0^2 q^2 R^2}{4m} t^5$

Figure shows a movable, rigid and conducting connector MN between a fixed L-shaped conducting wire. The whole system is in horizontal plane having vertical magnetic field B and point M is pulled horizontally with constant speed v starting from t = 0 by some external agent. Initially, at t = 0, if $\theta = 90°$, induced current in the loop OMN is found to be zero at time $t = \frac{\ell}{\sqrt{K V}}$.. Find K (Given that M and N are always in contact with the L–shaped wire).



A vessel of volume V_0 is evacuated by means of a piston air pump. One piston stroke captures the volume $\Delta V = 0.2\ V_0$. If process is assumed to be isothermal then find the minimum number of strokes after which pressure in the vessel becomes $\left(\frac{1}{1.728}\right)$ $(P_{initial})$.

The current density $\vec{J}$ inside a long, solid, cylindrical wire of radius a = 12 mm is in the direction of the central axis, and its magnitude varies linearly with radial distance r from the axis according $J = \frac{J_0 r}{a}$, , where $J_0 = \frac{10^5}{4\pi} A/m^2$. Find the magnitude of the magnetic field at $r = \frac{a}{2}$ in μT.

A locomotive wheel is to be fitted with a steel band. The 21 kg steel band of diameter lm has a temperature of 20°C and diameter 6×10^{-4} m less than that of the wheel. It is proposed to condense some steam at 100°C on the steel band so that it can fit on the wheel. What is the mass of steam required (in gm). $S_{steel} = 570 J/kg°C, S_{wate} = 4200 J/kg°C, a_{steel} = 12 \times 10^{-6}/°C, L_{team} = 540$ cal/gm.

A planoconvex lens has a thickness of = 4 cm. When placed on a horizontal surface (table), with the covered surface in contact with it, the apparent depth of the bottom most point of the lens is found to be t_1 = 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the center of the plane surface is found to be t_2 = 25/8 cm. Find the focal length of the lens (in cm).

A proton X is projected in a region of uniform electric field E with speed u. Another proton Y is projected simultaneously with same speed u in a separate region having a uniform magnetic field B. It is found that both X and Y has same debroglie wavelength after time T. The specific charge of proton is σ C/kg.

The magnitude of displacement of proton X in time T is found to $\left[\sqrt{u^2 - \frac{\sigma^2 E^2 T^2}{K}}\right]T$. The value of K is

The pitch of the screw gauge is 1mm. There are 100 divisions on the circular scale. When the gap between the jaws is just closed, 8^{th} division of the circular scale coincides with reference line. While measuring the diameter of a wire, the linear scale reads 1mm and 50^{th} division on the circular scale coincides with the reference line. The length of the wire is 8.6 cm. The curved surface area of the cylinder is_____(in cm^2).